

Digital Employee Platform

System Design — Text Reference

Complete reference for all 23 database entities, field types, constraints, entity relationships, and UML class structures. Use this document when recreating or extending the platform schema.

Contents

1. Database Schema — 23 Entities with all fields
2. Entity Relationships — cardinality table
3. UML Class Structures — properties + methods per class
4. Architecture Notes — Clean Architecture layers

Section 1 — Database Schema

Legend: **Gold** = Primary Key **Teal** = Foreign Key * = required / NOT NULL

1.1 Core Domain

Business

Field	Type	Constraint	Notes
Id	Guid	PK	Unique tenant identifier
Name	string	NOT NULL	Business display name
Type	string		e.g. Restaurant, Retail
Phone	string		Contact phone number
Email	string	UNIQUE	Business email
City	string		City / location
IsActive	bool	NOT NULL	Soft-enable/disable tenant
CreatedAt	DateTime	NOT NULL	UTC timestamp

User

Field	Type	Constraint	Notes
Id	Guid	PK	Staff member identifier
BusinessId	Guid	FK	→ Business.Id
FullName	string	NOT NULL	
Email	string	UNIQUE	Login email
PasswordHash	string	NOT NULL	BCrypt hash
Role	string	NOT NULL	Admin Staff Manager
IsActive	bool	NOT NULL	
CreatedAt	DateTime	NOT NULL	

Customer

Field	Type	Constraint	Notes
CustomerId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
FullName	string	NOT NULL	
Email	string		Optional contact
Phone	string		Optional contact
CreatedAt	DateTime	NOT NULL	

Setting

Field	Type	Constraint	Notes
SettingId	Guid	PK	
BusinessId	Guid	FK, UNIQUE	→ Business.Id (1-to-1)
ChatbotEnabled	bool	NOT NULL	Toggle AI chatbot
AgentVoice	string		TTS voice name
Language	string		Default language code
CustomSystemPrompt	string		Override LLM system prompt
WelcomeMessage	string		

WorkingHours

Field	Type	Constraint	Notes
WorkingHoursId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
DayOfWeek	int	NOT NULL	0=Sun ... 6=Sat
OpenTime	TimeOnly	NOT NULL	Opening time
CloseTime	TimeOnly	NOT NULL	Closing time
IsClosed	bool	NOT NULL	True = closed all day

1.2 Chat & AI Domain

Interaction

Field	Type	Constraint	Notes
InteractionId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
CustomerId	Guid	FK	→ Customer.CustomerId
Channel	string	NOT NULL	Chat Voice WhatsApp Web
Status	string	NOT NULL	Active Ended Escalated
StartedAt	DateTime	NOT NULL	
EndedAt	DateTime		Null if still active

Message

Field	Type	Constraint	Notes
MessageId	Guid	PK	
InteractionId	Guid	FK	→ Interaction.InteractionId
Content	string	NOT NULL	Text content
SenderType	string	NOT NULL	Customer AI Agent
SentAt	DateTime	NOT NULL	
TokensUsed	int		LLM token count

Sentiment

Field	Type	Constraint	Notes
SentimentId	Guid	PK	
MessageId	Guid	FK, UNIQUE	→ Message.MessageId (1-to-1)
Score	float	NOT NULL	-1.0 (neg) to +1.0 (pos)
Label	string	NOT NULL	Positive Neutral Negative
AnalyzedAt	DateTime	NOT NULL	

InteractionAnalysis

Field	Type	Constraint	Notes
InteractionAnalysisId	Guid	PK	
InteractionId	Guid	FK, UNIQUE	→ Interaction (1-to-1)
MainIntent	string		e.g. OrderInquiry, Complaint
SentimentScore	float		Average across messages
TopicsJson	string		JSON array of key topics
AnalyzedAt	DateTime	NOT NULL	

CallSummary

Field	Type	Constraint	Notes
Id	Guid	PK	
InteractionId	Guid	FK, UNIQUE	→ Interaction (1-to-1)
Summary	string	NOT NULL	AI-generated summary
KeyTopics	string		Comma-separated topics
ActionItems	string		Follow-up actions
GeneratedAt	DateTime	NOT NULL	

1.3 Orders & Menu Domain

MenuCategory

Field	Type	Constraint	Notes
MenuCategoryId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
Name	string	NOT NULL	e.g. Starters, Mains, Desserts
SortOrder	int		Display ordering
IsActive	bool	NOT NULL	

MenuItem

Field	Type	Constraint	Notes
MenuItemId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
MenuCategoryId	Guid	FK	→ MenuCategory.MenuCategoryId
Name	string	NOT NULL	
Description	string		
Price	decimal	NOT NULL	
IsAvailable	bool	NOT NULL	
ImageUrl	string		

Order

Field	Type	Constraint	Notes
OrderId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
CustomerId	Guid	FK	→ Customer.CustomerId
TotalPrice	decimal	NOT NULL	
Status	string	NOT NULL	Pending Confirmed Delivered Cancelled
Notes	string		Customer notes
CreatedAt	DateTime	NOT NULL	
UpdatedAt	DateTime		

OrderItem

Field	Type	Constraint	Notes
OrderItemId	Guid	PK	
OrderId	Guid	FK	→ Order.OrderId
MenuItemId	Guid	FK	→ MenuItem.MenuItemId
Quantity	int	NOT NULL	
UnitPrice	decimal	NOT NULL	Price at time of order
Notes	string		

1.4 Support & Notifications Domain

Ticket

Field	Type	Constraint	Notes
Id	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
CustomerId	Guid	FK	→ Customer.CustomerId
Subject	string	NOT NULL	
Description	string		
Status	string	NOT NULL	Open InProgress Resolved Closed
PriorityLevel	string	NOT NULL	Low Medium High Critical
AssignedToUserId	Guid	FK	→ User.Id (nullable)
CreatedAt	DateTime	NOT NULL	
ResolvedAt	DateTime		

Feedback

Field	Type	Constraint	Notes
FeedbackId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
CustomerId	Guid	FK	→ Customer.CustomerId
TicketId	Guid	FK	→ Ticket.Id (nullable)
Rating	int	NOT NULL	1–5 stars
Comment	string		
SentimentScore	float		AI-analysed score
SubmittedAt	DateTime	NOT NULL	

Notification

Field	Type	Constraint	Notes
NotificationId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
UserId	Guid	FK	→ User.Id (nullable)
Title	string	NOT NULL	
Body	string		
Type	string	NOT NULL	Info Warning Alert
IsRead	bool	NOT NULL	
CreatedAt	DateTime	NOT NULL	

1.5 Business Management Domain

Subscription

Field	Type	Constraint	Notes
Id	Guid	PK	
BusinessId	Guid	FK, UNIQUE	→ Business.Id
PlanName	string	NOT NULL	Free Pro Enterprise
Price	decimal	NOT NULL	Monthly price
StartDate	DateTime	NOT NULL	
EndDate	DateTime		Null = ongoing
IsActive	bool	NOT NULL	

PaymentTransaction

Field	Type	Constraint	Notes
Id	Guid	PK	
SubscriptionId	Guid	FK	→ Subscription.Id
Amount	decimal	NOT NULL	
Currency	string	NOT NULL	Default: USD
PaymentMethod	string	NOT NULL	Card PayPal BankTransfer
Status	string	NOT NULL	Pending Completed Failed Refunded
TransactionRef	string		External payment gateway ref
PaidAt	DateTime	NOT NULL	

KnowledgeBase

Field	Type	Constraint	Notes
KnowledgeBaseld	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
Question	string	NOT NULL	
Answer	string	NOT NULL	
Category	string		Topic grouping
IsFAQ	bool	NOT NULL	Show in FAQ widget
IsActive	bool	NOT NULL	
CreatedAt	DateTime	NOT NULL	

Integration

Field	Type	Constraint	Notes
Id	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
PlatformName	string	NOT NULL	WhatsApp Slack Zapier etc.
ApiKeyOrConfig	string	NOT NULL	Encrypted JSON config
Status	string	NOT NULL	Active Inactive Error
ConnectedAt	DateTime	NOT NULL	

AuditLog

Field	Type	Constraint	Notes
AuditLogId	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
UserId	Guid	FK	→ User.Id (nullable)
Action	string	NOT NULL	e.g. Create, Update, Delete
Entity	string	NOT NULL	Table / resource name
EntityId	string		Affected record id
OldValue	string		JSON snapshot before
NewValue	string		JSON snapshot after
Timestamp	DateTime	NOT NULL	
IpAddress	string		

Report

Field	Type	Constraint	Notes
Id	Guid	PK	
BusinessId	Guid	FK	→ Business.Id
Title	string	NOT NULL	
ReportType	string	NOT NULL	Sales Support Chat Custom
Parameters	string		JSON filter parameters
FilePath	string		Generated file path
GeneratedAt	DateTime	NOT NULL	
GeneratedBy	Guid	FK	→ User.Id

Section 2 — Entity Relationships

All foreign-key relationships with cardinality.

Entity A	Cardinality	Entity B	Relationship	Notes
Business	1 → N	User	EMPLOYS	Multi-tenant staff
Business	1 → N	Customer	SERVES	Per-tenant customers
Business	1 → 1	Setting	CONFIGURES	One config per tenant
Business	1 → N	WorkingHours	HAS_HOURS	7 rows (one per day)
Business	1 → N	Interaction	TRACKS	
Business	1 → N	Order	RECEIVES	
Business	1 → N	Ticket	MANAGES	
Business	1 → N	Notification	SENDS	
Business	1 → 1	Subscription	HAS	
Business	1 → N	MenuItem	OFFERS	
Business	1 → N	MenuCategory	HAS	
Business	1 → N	KnowledgeBase	OWNS	
Business	1 → N	Integration	USES	
Business	1 → N	AuditLog	LOGS	
Business	1 → N	Report	GENERATES	
Customer	1 → N	Interaction	INITIATES	
Customer	1 → N	Order	PLACES	
Customer	1 → N	Ticket	RAISES	
Customer	1 → N	Feedback	GIVES	
Interaction	1 → N	Message	CONTAINS	
Interaction	1 → 1	InteractionAnalysis	ANALYZED_AS	
Interaction	1 → 1	CallSummary	SUMMARIZED	
Message	1 → 1	Sentiment	SCORED_BY	
MenuCategory	1 → N	MenuItem	GROUPS	
Order	1 → N	OrderItem	CONTAINS	
MenuItem	1 → N	OrderItem	USED_IN	
Ticket	1 → N	Feedback	RECEIVES	
Subscription	1 → N	PaymentTransaction	PAID_VIA	
User	1 → N	AuditLog	LOGGED_BY	nullable
User	1 → N	Report	GENERATED_BY	

Section 3 — UML Class Structures

Each class lists its C# properties and key methods. Visibility: + public, - private, # protected.

3.1 Core Domain Classes

Business	User
— Properties —	— Properties —
+ Id: Guid	+ Id: Guid
+ Name: string	+ BusinessId: Guid
+ Type: string	+ FullName: string
+ IsActive: bool	+ Email: string
+ CreatedAt: DateTime	+ PasswordHash: string
+ Users: ICollection	+ Role: UserRole (enum)
+ Customers: ICollection	— Methods —
+ Setting: Setting	+ SetPassword(plain: string): void
— Methods —	+ VerifyPassword(plain: string): bool
+ Activate(): void	+ HasRole(role: UserRole): bool
+ Deactivate(): void	
+ GetActiveUsers(): IList	

Customer	Setting
— Properties —	— Properties —
+ CustomerId: Guid	+ SettingId: Guid
+ BusinessId: Guid	+ BusinessId: Guid
+ FullName: string	+ ChatbotEnabled: bool
+ Email: string	+ AgentVoice: string
+ Phone: string	+ CustomSystemPrompt: string
— Methods —	+ WelcomeMessage: string
+ GetOrderHistory(): IList	— Methods —
+ GetActiveInteraction(): Interaction?	+ BuildSystemPrompt(): string
	+ Toggle(feature: string): void

3.2 Chat & AI Domain Classes

Interaction	Message
— Properties —	— Properties —
+ InteractionId: Guid	+ MessageId: Guid
+ BusinessId: Guid	+ InteractionId: Guid
+ CustomerId: Guid	+ Content: string
+ Channel: ChannelType	+ SenderType: SenderType
+ Status: InteractionStatus	+ SentAt: DateTime
+ Messages: ICollection	+ Sentiment: Sentiment?
— Methods —	— Methods —
+ Start(): void	+ IsFromCustomer(): bool
+ End(): void	+ IsFromAI(): bool
+ Escalate(): void	
+ GetDuration(): TimeSpan	

Sentiment
— Properties —
+ SentimentId: Guid
+ MessageId: Guid
+ Score: float
+ Label: SentimentLabel
— Methods —
+ IsNegative(): bool
+ IsPositive(): bool

InteractionAnalysis
— Properties —
+ InteractionAnalysisId: Guid
+ InteractionId: Guid
+ MainIntent: string
+ SentimentScore: float
+ TopicsJson: string
— Methods —
+ GetTopics(): List
+ OverallSentiment(): SentimentLabel

3.3 Orders & Menu Classes

Order
— Properties —
+ OrderId: Guid
+ BusinessId: Guid
+ CustomerId: Guid
+ Status: OrderStatus
+ TotalPrice: decimal
+ Items: ICollection
— Methods —
+ Confirm(): void
+ Cancel(): void
+ CalculateTotal(): decimal
+ AddItem(item: OrderItem): void

MenuItem
— Properties —
+ MenuItemId: Guid
+ BusinessId: Guid
+ MenuCategoryId: Guid
+ Name: string
+ Price: decimal
+ IsAvailable: bool
— Methods —
+ ToggleAvailability(): void
+ UpdatePrice(p: decimal): void

3.4 Support & Notifications Classes

Ticket
— Properties —
+ Id: Guid
+ BusinessId: Guid
+ CustomerId: Guid
+ Subject: string
+ Status: TicketStatus
+ PriorityLevel: Priority
+ AssignedToUserId: Guid?
— Methods —
+ Assign(userId: Guid): void
+ Resolve(): void
+ Escalate(): void
+ Reopen(): void

Notification
— Properties —
+ NotificationId: Guid
+ BusinessId: Guid
+ UserId: Guid?
+ Title: string
+ Type: NotificationType
+ IsRead: bool
— Methods —
+ MarkAsRead(): void
+ IsForUser(id: Guid): bool

3.5 Management Classes

Subscription
— Properties —
+ Id: Guid
+ BusinessId: Guid
+ PlanName: string
+ Price: decimal
+ IsActive: bool
+ Transactions: ICollection
— Methods —
+ Upgrade(plan: string): void
+ Cancel(): void
+ IsExpired(): bool

KnowledgeBase
— Properties —
+ KnowledgeBaseId: Guid
+ BusinessId: Guid
+ Question: string
+ Answer: string
+ IsFAQ: bool
+ IsActive: bool
— Methods —
+ Activate(): void
+ Deactivate(): void

AuditLog
— Properties —
+ AuditLogId: Guid
+ BusinessId: Guid
+ UserId: Guid?
+ Action: string
+ Entity: string
+ OldValue: string?
+ NewValue: string?
+ Timestamp: DateTime
— Methods —
static Log(action, entity, ...): AuditLog

Integration
— Properties —
+ Id: Guid
+ BusinessId: Guid
+ PlatformName: string
+ ApiKeyOrConfig: string
+ Status: IntegrationStatus
— Methods —
+ Connect(): Task
+ Disconnect(): void
+ Test(): Task

Section 4 — Architecture Notes

Pattern: Clean Architecture (.NET 9 / ASP.NET Core 9)

Layer	Project	Responsibility	Key Contents
Domain	Domain	Business rules, no dependencies	<i>Entities, Enums, Domain Events, Value Objects</i>
Application	Application	Use cases, orchestration	<i>Commands/Queries (CQRS), DTOs, Interfaces, MediatR handlers</i>
Infrastructure	Infrastructure	External concerns	<i>EF Core DbContext, Repositories, AI services, Email, JWT</i>
API	API	HTTP entry point	<i>Controllers, Middleware, Filters, Program.cs, appsettings</i>

Key Technology Choices

- ORM: Entity Framework Core 9 (Code-First, Migrations)
- Auth: ASP.NET Core Identity + JWT Bearer tokens
- CQRS: MediatR (Commands + Queries separated)
- Validation: FluentValidation on all Commands/Queries
- AI: OpenAI / Azure OpenAI SDK for chat, sentiment, summarisation
- Real-time: SignalR for live chat & notifications
- Background Jobs: Hangfire for scheduled reports & digests
- API Docs: Swagger / OpenAPI auto-generated
- Multi-tenancy: BusinessId on every entity (row-level isolation)